

PROBLEM SET 3—SOLUTIONS

ECON301—ECONOMETRICS I

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These are my suggested solutions. It is your responsibility to cross-check this document with your own answers.

1. Define Z_n as:

$$Z_n = \frac{\frac{1}{n} \sum_{i=1}^n W_i}{\sigma_W / \sqrt{n}} = \frac{\frac{1}{\sqrt{n}} \sum_{i=1}^n W_i}{\sigma_W}.$$

Since $E(W_i) = 0$ for all i , by the CLT we have that

$$Z_n = \frac{\frac{1}{\sqrt{n}} \sum_{i=1}^n W_i}{\sigma_W} \rightarrow_d N(0, 1).$$

Using Slutsky's Theorem (Thm. 4.6, ii) we have then that

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n W_i \rightarrow_d \sigma_W N(0, 1) = N(0, \sigma_W^2).$$

An alternative derivation:

$$\Pr\left(\frac{1}{\sqrt{n}} \sum_{i=1}^n W_i \leq w\right) = \Pr\left(\frac{1}{n} \sum_{i=1}^n W_i \leq \frac{w}{\sqrt{n}}\right) = \Pr\left(Z_n \leq \frac{w/\sqrt{n}}{\sigma_W/\sqrt{n}}\right) = \Pr(Z_n \leq w/\sigma_W) \approx \Phi(w/\sigma_W).$$

Therefore,

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n W_i \overset{A}{\sim} N(0, \sigma_W^2).$$

2. We will first show that $X_n \rightarrow_{L_2} c$, which by theorem 4.1 implies that $X_n \rightarrow_P c$. We have that

$$\begin{aligned} E[(X_n - c)^2] &= E[(X_n - E(X_n) + E(X_n) - c)^2] = E[(X_n - E(X_n))^2 + 2(X_n - E(X_n))(E(X_n) - c) + (E(X_n) - c)^2] \\ &= E[(X_n - E(X_n))^2] + 2(E(X_n) - E(X_n))(E(X_n) - c) + (E(X_n) - c)^2 \\ &= \text{Var}(X_n) + (E(X_n) - c)^2 \end{aligned}$$

Taking limits as n tends to infinity and using the assumption that $\lim_{n \rightarrow \infty} E(X_n) = c$ we have that $E[(X_n - c)^2] \rightarrow 0$. Therefore, we conclude that $X_n \rightarrow_{L_2} c$ and by theorem 4.1, the convergence is also in probability.

3. Let $g(X) = (X - \mu)^2 \geq 0$. Then apply Chebychev's inequality (as in the lecture) with $r = k^2 \sigma^2$ to get

$$\Pr(|X - \mu| \geq k\sigma) = \Pr((X - \mu)^2 \geq k^2 \sigma^2) \leq \frac{E[(X - \mu)^2]}{k^2 \sigma^2} = \frac{\sigma^2}{k^2 \sigma^2} = \frac{1}{k^2}$$

4. We know that $E[\bar{X}] = \mu$ and $\text{Var}(\bar{X}) = \sigma_{\bar{X}}^2 = \sigma^2/n = 9/100$ so that $\sigma_{\bar{X}} = \sqrt{9/100} = 0.3$.

(a) Using Chebychev's second inequality, we get

$$\Pr(-0.3k < \bar{X} - \mu < 0.3k) \geq 1 - 1/k^2.$$

We need $1 - 1/k^2 \geq 0.9$ which implies that $k \geq \sqrt{10} = 3.1623$. Thus, we can take $k = 3.1623$. This implies that

$$\Pr(-0.9487 < \bar{X} - \mu < 0.9487) \geq 0.9.$$

(b) Using the CLT, $\bar{X} \sim AN(\mu, \sigma_{\bar{X}}^2)$. and $(\bar{X} - \mu)/0.3 \sim AN(0, 1)$. Thus

$$\Pr\left(-1.645 < \frac{\bar{X} - \mu}{0.3} < 1.645\right) = \Pr(-0.4935 < \bar{X} - \mu < 0.4935) \approx 0.9.$$

Thus, the limits obtained by the CLT are tighter than the ones obtained by Chebychev's Second Inequality. In fact, the ones obtained by Chebychev's are almost twice as big as the ones normal approximation. Moreover, with a sample size of 100, $\bar{X}$ is probably very close to normally distributed, even if the underlying X distribution is not close to normal. On the other hand, one reason that Chebychev's Inequality is widely applicable to any positive random variable with finite mean and variance is that the assumptions required to use it are quite mild, but the trade-off for this wide applicability is that the bounds obtained with it are quite conservative, i.e., not very tight.

5. (a) Since $E(Y_i) = p$ and $\text{Var}(Y_i) = p(1-p)$ and we have a random sample, the conditions of the WLLN (thm 4.4) apply so we have that $\hat{p}_n \rightarrow_P E(Y_i) = p$.
- (b) Recall that if Y_i is a Bernoulli random variable with parameter p , we have that $E(Y_i) = p$ and $\text{Var}(Y_i) = p(1-p)$. Therefore, since $\{Y_i\}$ is a sequence of i.i.d. random variables, $E(\hat{p}) = p$ and $\text{Var}(\hat{p}) = p(1-p)/n$. Therefore, the sequence

$$Z_n = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}}$$

satisfies the conditions of the CLT and we can conclude that $Z_n \rightarrow_d N(0, 1)$.

- (c) We propose the following estimator for $\text{Var}(Y_1)$:

$$\widehat{\text{Var}}(Y_1) = \hat{p}(1 - \hat{p}).$$

By the WLLN, we know that $\hat{p} \rightarrow_P p$. Define $g(y) = y(1 - y)$. Then g is a continuous function in all of its domain and in light of this, we can apply theorem 4.3 to conclude that

$$\widehat{\text{Var}}(Y_1) = \hat{p}(1 - \hat{p}) = g(\hat{p}) \rightarrow_P g(p) = p(1 - p) = \text{Var}(Y_1).$$

Therefore, we $\widehat{\text{Var}}(Y_1)$ is a consistent estimator.

- (d) Let T_n be defined as

$$T_n = \frac{\hat{p} - p}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}},$$

and let Z_n be defined as in item (a). Firstly, note that $\sqrt{\hat{p}(1-\hat{p})} \rightarrow_P \sqrt{p(1-p)}$, i.e. $\sqrt{\widehat{\text{Var}}(Y_1)}$ is a consistent estimator of $\sqrt{p(1-p)}$ where, once again, this was an application of theorem 4.3. Define W_n as

$$W_n = \frac{\sqrt{\hat{p}(1-\hat{p})}}{\sqrt{p(1-p)}}.$$

Then we have that

$$T_n = \frac{\hat{p} - p}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}} = \frac{\hat{p} - p}{\frac{\sqrt{p(1-p)}}{\sqrt{n}} \frac{\sqrt{\hat{p}(1-\hat{p})}}{\sqrt{p(1-p)}}} = \frac{Z_n}{W_n},$$

since $W_n \rightarrow_P 1 \neq 0$, we have from theorem 4.6 (iii)

$$T_n \rightarrow_d \frac{N(0, 1)}{1} = N(0, 1).$$

Thus, T_n converges in distribution to a standard normal distribution when n tends to infinity.

- (e) Firstly, we have that

$$\hat{p} = \frac{39}{127} \approx 0.3071, \quad \sqrt{\frac{\hat{p}(1-\hat{p})}{127}} \approx 0.0409.$$

The large sample approximation for the confidence interval is

$$\Pr(|T_n| \leq z_{\alpha/2}) = \Pr\left(\left|\frac{\hat{p} - p}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}}\right| \leq z_{\alpha/2}\right) \approx 1 - \alpha.$$

After some simple algebraic manipulations, we obtain:

$$\Pr\left(\hat{p} - z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \leq p \leq \hat{p} + z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}\right) \approx 1 - \alpha.$$

For a 99% confidence interval, $\alpha = 0.01$ so that $z_{0.005} = 2.57583$. Thus, the 99% confidence interval from the large sample approximation is given by

$$[0.3071 \pm 2.57583(0.0409)] = [0.2017, 0.4125].$$

The above is a random interval that will have an *approximate* probability of 99% of containing the true value of p .

- (f) Let $\hat{\gamma}_n = \hat{p}_n/(1 - \hat{p}_n)$ and define $g(x) = x/(1 - x)$; g is continuous for all $x \in (0, 1)$, and assuming that $p \in (0, 1)$, we can use theorem 4.3 to get $\hat{\gamma}_n = g(\hat{p}_n) = \hat{p}_n/(1 - \hat{p}_n) \rightarrow_P g(p) = p/(1 - p) = \gamma$.
- (g) We know from (a) that: $\sqrt{n}(\hat{p}_n - p) \rightarrow_d N(0, p(1 - p))$ and $\hat{\gamma}_n = g(\hat{p}_n)$; since g is continuously differentiable on the interval $(0, 1)$ and $g'(x) = 1/(1 - x)^2$ as you can check, we can use the Delta Theorem as stated to get

$$\sqrt{n}(\hat{\gamma}_n - \gamma) \rightarrow_d N(0, p(1 - p)/(1 - p)^4) = N(0, p/(1 - p)^3).$$

- (h) Let $s_n^2 = \frac{\hat{p}_n}{(1 - \hat{p}_n)^3}$, since $\hat{\gamma}_n \rightarrow_P \gamma$, theorem 4.3 applies: define $h(x) = x/(1 - x)^3$, h is continuous on the interval $(0, 1)$, so we have that $s_n^2 = h(\hat{p}_n) \rightarrow_P h(p) = p/(1 - p)^3$.
- (i) We are going to use the same technique as in (d) to construct the asymptotic pivot for γ . Define

$$Z_n = \frac{\hat{\gamma}_n - \gamma}{\sqrt{\frac{p}{n(1-p)^3}}},$$

from item (g) we know that $Z_n \rightarrow_d N(0, 1)$. Applying theorem 4.3 and item (h) yields $s_n = \sqrt{s_n^2} \rightarrow_P \sqrt{\frac{p}{1-p}^3}$. Then we have that:

$$W_n = \frac{\sqrt{s_n^2}}{\sqrt{\frac{p}{(1-p)^3}}} \rightarrow_P \frac{\sqrt{\frac{p}{(1-p)^3}}}{\sqrt{\frac{p}{(1-p)^3}}} = 1 \neq 0,$$

where the last part was an application of theorem 4.3. Now define $T_n = Z_n/W_n$ to get

$$T_n = \frac{Z_n}{W_n} = \frac{\frac{\hat{\gamma}_n - \gamma}{\sqrt{\frac{p}{n(1-p)^3}}}}{\frac{\sqrt{s_n^2}}{\sqrt{\frac{p}{(1-p)^3}}}} = \frac{\hat{\gamma}_n - \gamma}{\sqrt{\frac{s_n^2}{n}}} = \frac{\hat{\gamma}_n - \gamma}{\frac{s_n}{\sqrt{n}}},$$

by Slutsky's Theorem (thm. 4.6 part iii) we have that $T_n \rightarrow_d N(0, 1)/1 = N(0, 1)$. Then T_n is the asymptotic pivot because all of its components can be estimated.

- (j) Let $z_{\alpha/2}$ be the $(1 - \alpha)\%$ critical value for from the standard normal distribution, then

$$\Pr(|T_n| \leq z_{\alpha/2}) \approx 1 - \alpha \iff \Pr(\hat{\gamma}_n - z_{\alpha/2} \cdot s_n \leq \gamma \leq \hat{\gamma}_n + z_{\alpha/2} \cdot s_n) \approx 1 - \alpha.$$

6. Note that $f(z; \theta)$ is actually an exponential distribution, which is a special case of the Gamma distribution with parameters $\text{Gamma}(1, 1/\theta)$. Using the results from the first homework, we have that $E[X_i] = \theta$ and $\text{Var}(X_i) = \theta^2$ for all i , as you can readily check by computing them. Also note that if we define the sample average as $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$, we have that $Y_n = \frac{1}{n} + \bar{X}_n$. Also recall that $E[\bar{X}_n] = \theta$ and $\text{Var}(\bar{X}_n) = \theta^2/n$.

- (a) Note that $E[Y_n] = 1/n + E[\bar{X}_n] = 1/n + \theta$, and $\text{Var}(Y_n) = \text{Var}(\bar{X}_n) = \theta^2/n$. Note that the proof of item 2 in this problem set (which is the proof of Corollary 4.2 in the lectures) implies that since $\lim_{n \rightarrow \infty} (1/n + \theta) = \theta$, and $\lim_{n \rightarrow \infty} \theta^2/n = 0$, we have that $Y_n \rightarrow_{L_2} \theta$, and therefore $Y_n \rightarrow_P \theta$.

- (b) By the CLT, we have that

$$\frac{\bar{X}_n - \theta}{\frac{\theta}{\sqrt{n}}} \rightarrow_d N(0, 1),$$

or equivalently, $\bar{X}_n \sim AN(\theta, \theta^2/n)$. Using Slutsky's Theorem, we have that $Y_n = 1/n + \bar{X}_n \sim AN(\theta + 1/n, \theta^2/n)$.

- (c) We have that $\Pr(Y_n \geq 15) = \Pr\left(\frac{Y_n - \theta - 1/n}{\theta/\sqrt{n}} \geq \frac{15 - \theta - 1/n}{\theta/\sqrt{n}}\right) \approx 1 - \Phi\left(\frac{15 - \theta - 1/n}{\theta/\sqrt{n}}\right)$, where Φ is the CDF of the standard normal distribution. Now, when $\theta = 10$, and $n = 100$, we have that the above probability can be approximated by

$$1 - \Phi(4.99) = 3.01896 \times 10^{-7} \approx 0$$

7. Note, as you can check by yourselves, that $E(X_i) = 1/2$ and $\text{Var}(X_i) = 1/12$ for all $i = 1, \dots, n$.

(a) We have that $E(\bar{X}_n) = 1/2$ and $\text{Var}(\bar{X}_n) = \frac{1}{12n}$. Define Z_n as

$$Z_n = \frac{\bar{X}_n - 1/2}{\sqrt{\frac{1}{12n}}} = \sqrt{12n} \bar{X}_n - \frac{\sqrt{12n}}{2} = \sqrt{12n} \left(n^{-1} \sum_{i=1}^n X_i \right) - \sqrt{3n} = \sqrt{\frac{12}{n}} \left(\sum_{i=1}^n X_i \right) - \sqrt{3n}$$

By the CLT, we have that $Z_n = \frac{\bar{X}_n - 1/2}{\sqrt{\frac{1}{12n}}} = \sqrt{\frac{12}{n}} \left(\sum_{i=1}^n X_i \right) - \sqrt{3n} \rightarrow_d N(0, 1)$. Thus, we can see that $\bar{X}_n \approx AN(1/2, 1/(12n))$.

(b) From the item above, we know that $Z_n \rightarrow_d N(0, 1)$. When $n = 12$, we have that $Z_{12} = \sqrt{\frac{12}{12}} \left(\sum_{i=1}^{12} X_i \right) - \sqrt{3 \cdot 12} = \left(\sum_{i=1}^{12} X_i \right) - 6$. Thus $\left(\sum_{i=1}^{12} X_i \right) - 6$ can be approximated by a standard normal distribution $N(0, 1)$. Even though $n = 12$ is quite the small sample, the approximation is actually quite good.